

Engineering Mechanics MEL 1010

Dynamics of particles Quiz-3

Total Time: 1 hr 15 minutes

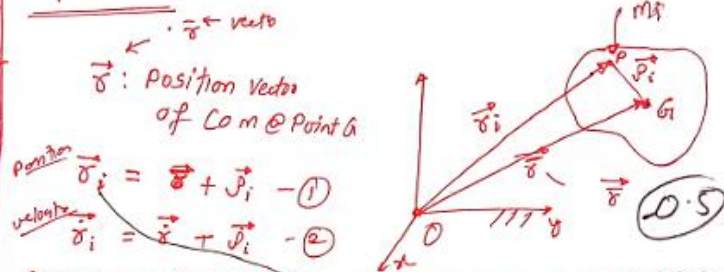
Total Marks: 60

Draw relevant figures. No extra time will be given. If you complete the quiz early, submit your answer sheet and leave the examination room quietly.

1. What do you understand by term 'angular momentum'. Marks:2
2. What do you understand by 'absolute angular momentum' and 'relative angular momentum'. Marks:4
3. Show that for a system of n particles, absolute angular momentum about center of mass is same as its relative angular momentum about center of mass. Marks:5
4. Derive Newton's second law of motion of a system of n particles.
Hint: 1. Write Newton's second law of motion of one particle considering external and internal forces. 2. Sum expression obtained in step 1 for all the n particles. Marks:5
5. Write work energy principle for a system of n particles. Does work done by internal forces need to be considered while applying work-energy principle for a system of n particles? Marks:5
6. Show that for a system of n particles, total kinetic energy of the system can be written in terms of velocity of center of mass and relative velocities of all the particles with respect to center of mass. Marks:5
7. Derive expression, for a system of n particles, relating rate of change of angular momentum about origin of a fixed reference system and moment of all the forces acting on all the particles about the same point. Do moments by internal forces need to be considered while calculating the total moment of all the forces about the fixed point? Marks:5
8. Show that for a system of n particles, rate of change of angular momentum about center of mass is equal to moment of all the forces acting on all the particles about center of mass. Do moments by internal forces need to be considered while calculating the total moment of all the forces about the center of mass? Marks:5
9. Show that for a system of n particles, rate of change of angular momentum about an arbitrary point P is not equal to moment of all the forces acting on all the particles about the same point. Marks:8
10. Derive the expression, for a system of n particles, between moment of all the forces about any arbitrary point P and rate of change of relative angular momentum about P . Marks:8
11. Show that for a system of n particles, absolute angular momentum about any arbitrary point P equals the angular momentum about G (center of mass) plus the moment about P of the linear momentum of the system considered concentrated at G . Marks:8

Note : Most of the Solution of Quiz 03 is in the lecture slides. {You can refer those}

Q3 Show that for a system of n particles, absolute angular momentum about center of mass is ~~same~~ same as its relative angular momentum about center of mass.



$\vec{r}$ ← vels
 $\vec{r}$: Position Vector of CoM @ Point G

Position $\vec{r}_i = \vec{r} + \vec{p}_i$ — (1)

velocity $\dot{\vec{r}}_i = \dot{\vec{r}} + \dot{\vec{p}}_i$ — (2)

Angular momentum about point G is sum of all angular moment of point P_i

$$H_G = \sum \vec{p}_i \times (m_i \dot{\vec{r}}_i) \quad \text{Linear momentum} \quad \text{--- (3)}$$

from (2)

$$= \sum \vec{p}_i \times m_i (\dot{\vec{r}} + \dot{\vec{p}}_i)$$

Moment of Linear momentum $\boxed{\vec{r} \times \vec{p}}$ Linear momentum

$$= \sum \vec{p}_i \times m_i \dot{\vec{r}} + \sum \vec{p}_i \times m_i \dot{\vec{p}}_i$$

$\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

$$= \sum -\dot{\vec{r}} \times (m_i \vec{p}_i) + \sum \vec{p}_i \times \frac{dm_i \vec{p}_i}{dt}$$

$$= \dot{\vec{r}} \times \sum m_i \vec{p}_i + \sum \vec{p}_i \times (m_i \dot{\vec{p}}_i)$$

show given Reason

$$= 0 + \sum \vec{p}_i \times m_i \dot{\vec{p}}_i \quad \text{relative velocity}$$

$$H_G = \sum \vec{p}_i \times (m_i \dot{\vec{p}}_i) = \text{Relative angular momentum}$$

relative Linear momentum about G

Sum of all

$$\therefore \bar{x} = \frac{\sum m_i x_i}{\sum m_i}$$

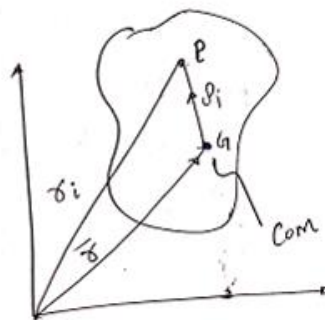
for CoM $\bar{x} = 0$ if $\sum m_i x_i = 0$
 But if $\sum m_i x_i \neq 0$
 Since $\vec{p}_i$ is position vector from CoM $\therefore \sum m_i \vec{p}_i = 0$

Part 1:

Show that for a system of n particles, rate of change of angular momentum about center of mass is equal to moment of all the forces acting on all the particles about center of mass.

$$H_G = \sum_i \underline{p}_i \times (m_i \dot{\underline{r}}_i)$$

$$\begin{aligned} \dot{H}_{G_1} &= \sum \dot{p}_i x(m_i, \dot{\sigma}_i) + \sum p_i x(m_i, \ddot{\sigma}_i) \\ &= \sum p_i x(m_i, \ddot{\sigma}_i) + \sum \dot{p}_i x(m_i, \dot{\sigma}_i) \end{aligned}$$



$$\text{II: } \sum_i \dot{p}_i \chi(m_i \dot{x}_i) = \sum_i (\dot{x}_i - \dot{\bar{x}}) \chi(m_i \dot{x}_i)$$

$$= \sum (\dot{x}_i \times \dot{x}_i) m_i - \sum m_i \dot{\bar{x}} \times \dot{x}_i$$

$$\sum m_i \delta \lambda \delta_i = 0$$

$$\dot{\vec{r}} \times \sum m_i (\dot{\vec{r}}_i + \dot{\vec{r}}) = 0 - \dot{\vec{r}} \times \sum_{i=1}^N m_i \dot{\vec{r}}_i$$

$$\dot{\vec{r}} \times \sum m_i \dot{\vec{r}}_i + \dot{\vec{r}} \times \sum m_i \dot{\vec{r}} = 0$$

$$\dot{\vec{r}} \times \sum_{i \neq 10} m_i \dot{\vec{r}}_i + \underbrace{\dot{\vec{r}} \times \sum m_i \dot{\vec{r}}_i}_{=0} = 0$$

$$0 - 0 = 0$$

$$\vec{r} = \frac{\sum m_i \vec{r}_i}{\sum m_i}$$

$$\dot{\vec{r}} = \frac{\sum m_i \dot{\vec{r}}_i}{\sum m_i}$$

velocity of C.M.

And $I: \sum p_i x(m; \ddot{x}_i)$

for Force acting on i th particles. $F_1 + F_2 + \dots + F_N + f_1 + f_2 + \dots + f_N = m_i \ddot{x}_i$

Thus $H_{G_0} = \sum_{i=1}^N p_i \times \left(\sum_{k=1}^K F_k + \sum_{\substack{j=1 \\ j \neq i}}^N f_{ji} \right)$ [all internal forces acting on i th particle]

Part 2

Do moments by internal forces need to be considered while calculating the total moment of all the forces about the center of mass?

$\vec{r}_i$: Position vector of i th particle

$\vec{r}_i$: Position vector of i th particle
 $\bar{\vec{r}}$: " ————— Com of System of particles

m_i : mass of i^{th} particle

m_i : mass of i^{th} particle
 M : Mass of system of particles, $\sum_{i=1}^N m_i = M$

N : Total no. of particles.

F_i : Force acting on i th particle

$$F_i = \sum_{j \neq i} F_{ij} + F_i^{\text{ext}} \quad \text{--- (1)}$$

F_{ij} : internal force

$$= \sum_{j \neq i} F_{ij} + F_i$$

F_{ij} $\rightarrow$ ~~Force~~ ^{Internal force} acting on i th Particle by j th particle. (Internal force)

F_i^{ext} : External force acting on i^{th} particle.

from proof (1) i.e. $H_G = \sum_{i=1}^N \rho_i \times (\sum_{j=i} F_{ij} + F_i^{ext})$ - (2)

Resultant (all internal forces)

↓ resultant
and External
forces acting on it.

~~Consider only the~~

~~Consider only the~~
Analysis of effect of internal forces

$$H_G = \underbrace{\sum_{i=1}^N p_i \times \left(\sum_{j \neq i}^N F_{ij} \right)}_{\text{III}} + \underbrace{\sum_{i=1}^N (p_i \times F_i^{\text{ext}})}_{\text{IV}}$$

$$\begin{aligned} \text{III} &= \sum_i p_i \times \left(\sum_{j \neq i}^N F_{ij} \right) \\ &= \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N (p_i \times F_{ij}) \quad \text{--- (3)} \end{aligned}$$

A/c to Newton's IIIrd Law $F_{ij} = -F_{ji} \quad \text{--- (4)}$

for each pair of particle.

$$\begin{aligned} (p_i \times F_{ij}) + (p_j \times F_{ji}) &= (p_i \times F_{ij}) - (p_j \times F_{ij}) \\ &= (p_i - p_j) \times (F_{ij}) \quad \text{--- (5)} \end{aligned}$$

Thus, eqn (3) can be written in pair wise as

$$\begin{aligned} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N (p_i \times F_{ij}) &= \sum_{i=1}^N \sum_{j=1}^N (p_i \times F_{ij} + p_j \times F_{ji}) \quad \text{Take sum of all pairs of} \\ &= 0 \quad \text{(from (5))} \end{aligned}$$

This proves that Moment by internal forces not need to be considered while calculating the total moment of all the forces about Com.



Note : Most of the Solution of Quiz 03 is in the lecture slides. {You can refer those}.

Quiz – 04

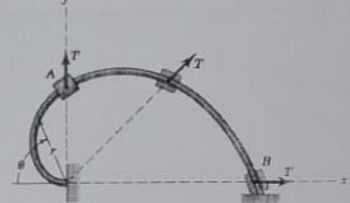
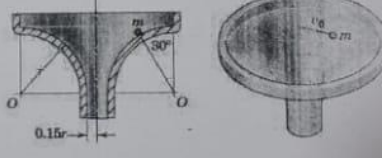
Engineering Mechanics MEL 1010 Dynamics of particles Quiz-4

Total Time: 1 hr 45 minutes

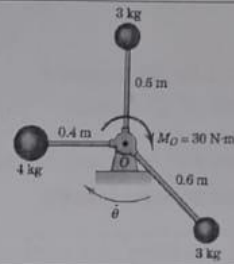
Total Marks: $10 \times 7 = 70$

Total Number of Questions: 7 [Each question carries 10 marks]

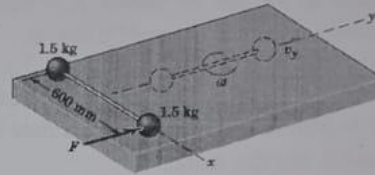
No extra time will be given. If you complete the quiz early, submit your answer sheet and leave the examination room quietly.

1. During a portion of a vertical loop, an airplane flies in an arc of radius $\rho = 600$ m with a constant speed $v = 400$ km/h. When the airplane is at A, the angle made by $\mathbf{v}$ with the horizontal is $\beta = 30^\circ$, and radar tracking gives $r = 800$ m and $\theta = 30^\circ$. Calculate v_r, v_θ, a_r, and $\ddot{\theta}$ for this instant. <i>10 marks</i>	
2. A ferry is moving due east and encounters a southwest wind of speed $v_w = 10$ m/s as shown. The experienced ferry captain wishes to minimize the effects of the wind on the passengers who are on the outdoor decks. At what speed v_B should he proceed? <i>10 marks</i>	
3. The 0.5-kg collar slides with negligible friction along the fixed spiral rod, which lies in the vertical plane. The rod has the shape of the spiral $r = 0.3\theta$, where r is in meters and θ is in radians. The collar is released from rest at A and slides to B under the action of a constant radial force $T = 10$ N. Calculate the velocity v of the slider as it reaches B. <i>10 marks</i>	
4. A particle is launched with a horizontal velocity $v_0 = 0.55$ m/s from the 30° position shown and then slides without friction along the funnel-like surface. Determine the angle θ which its velocity vector makes with the horizontal as the particle passes level $O-O$. The value of r is 0.9 m.	

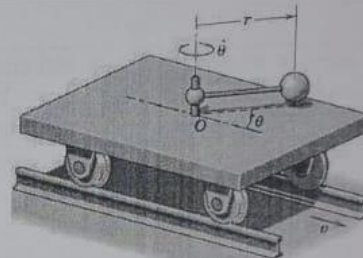
5. The three small spheres are welded to the light rigid frame which is rotating in a horizontal plane about a vertical axis through O with an angular velocity $\dot{\theta} = 20 \text{ rad/s}$. If a couple $M_O = 30 \text{ N}\cdot\text{m}$ is applied to the frame for 5 seconds, compute the new angular velocity $\dot{\theta}'$.



6. The two spheres are rigidly connected to the rod of negligible mass and are initially at rest on the smooth horizontal surface. A force F is suddenly applied to one sphere in the y -direction and imparts an impulse of $10 \text{ N}\cdot\text{s}$ during a negligibly short period of time. As the spheres pass the dashed position, calculate the velocity of each one.



7. The small car, which has a mass of 20 kg , rolls freely on the horizontal track and carries the 5-kg sphere mounted on the light rotating rod with $r = 0.4 \text{ m}$. A geared motor drive maintains a constant angular speed $\dot{\theta} = 4 \text{ rad/s}$ of the rod. If the car has a velocity $v = 0.6 \text{ m/s}$ when $\theta = 0$, calculate v when $\theta = 60^\circ$. Neglect the mass of wheels and any friction.



Solutions:

Q1

Q1.1:

$$V_u = V \cos(90 - \theta - \beta) = V \sin(\theta + \beta) = \dot{r}$$

$$V_\theta = V \sin(90 - \theta - \beta) = V \cos(\theta + \beta) = r\dot{\theta}$$

Since it moves with const. speed, it has only normal acceleration

$$a_n = \frac{V^2}{\rho}$$

$$a_u = a_n \cos(\theta + \beta) = \frac{V^2}{\rho} \cos(\theta + \beta) = \ddot{r} - r\dot{\theta}^2$$

$$a_\theta = -a_n \sin(\theta + \beta) = -\frac{V^2}{\rho} \sin(\theta + \beta) = 2\dot{r}\dot{\theta} + r\ddot{\theta}$$

For given instant, $\theta = 30^\circ$, $\beta = 30^\circ$, $V = 400 \text{ km/h} = 111.11 \text{ m/s}$

$$\therefore V_u = 96.22 \text{ m/s}$$

$$\therefore V_\theta = 55.55 \text{ m/s}$$

$$\therefore a_u = 10.28 \text{ m/s}^2$$

4 marks

Now, $a_\theta = -\frac{V^2}{\rho} \sin(\theta + \beta) = 2\dot{r}\dot{\theta} + r\ddot{\theta}$

$$= \frac{2V \sin(\theta + \beta)}{r} V \cos(\theta + \beta) + r\ddot{\theta}$$

$$\ddot{\theta} = \left(\frac{-V^2}{\rho} \sin(\theta + \beta) - \frac{V^2}{r} \sin(2(\theta + \beta)) \right) \frac{1}{r}$$

$$\ddot{\theta} /_{\theta=30^\circ, \beta=30^\circ} = -15.43179 \left(\frac{\sin 60^\circ}{600} + \frac{\sin 120^\circ}{800} \right)$$

$$\ddot{\theta} = -0.0389 \text{ rad/s}^2$$

Q2

Ques 2.

2 marks

$V_W = 10 \text{ m/s}$

From sin rule -

2 marks $\frac{V_W}{\sin \theta} = \frac{V_{W/B}}{\sin 50} = \frac{V_B}{\sin (180 - (50 + \theta))} \rightarrow \textcircled{1}$

from Eq. 1,

2 marks $\frac{V_W}{\sin \theta} = \frac{V_{W/B}}{\sin 50} \Rightarrow V_{W/B} = \frac{\sin 50 V_W}{\sin \theta}$

For minimization of the effects of the wind

2 marks $\Rightarrow V_{W/B} \rightarrow \text{minimum}$

$\Rightarrow \sin \theta \Rightarrow \text{maximum}$

$\Rightarrow \theta = 90^\circ$

Now from Eq. 1,

2 marks $\frac{V_W}{\sin (90)} = \frac{V_B}{\sin (180 - (50 + 90))} \Rightarrow V_B = 6.427 \text{ m/s}$

Q

Q3

Q3

Ans (8) mass = 0.5 kg $r = 0.30$ $\tau = 10 \text{ N}$
 cmf

$v_A = 0$
 $v_B = ?$
 Slides at B.

$\theta_1 = \pi/2$
 $\theta_2 = \pi/2$

Work-Energy Conservation

④ $\leftarrow K.E_{A-B} + P.E_{A-B} + \text{Work done}$

② $\leftarrow \frac{1}{2} m (v_A^2 - v_B^2) + mg(\Delta z) + f \cdot d z = 0$

② $\leftarrow \frac{1}{2} m (-v_B^2) + mg(0.30) + 10 \times 0.3 = 0$

② $\leftarrow \Rightarrow \frac{-v_B^2 \times 0.5}{2} + 0.5 \times 9.8 \times 0.3 \times (0) + 10 \times 0.3 (0.2 - 0.1)$

$\Rightarrow \frac{-v_B^2}{4} + \frac{9.8 \times 3}{20} \times \left(\frac{\pi}{2}\right) + 3 (\pi - \pi/2) = 0$

$\Rightarrow \frac{-v_B^2}{4} + 2.315 + 4.713 = 0$

$7.02 \times 4 = v_B^2$

$v_B = 5.3 \text{ m/sec}$

Wf 3.0
 Friction
 reprob 2

Q4

10 Marks

Given, $r = 0.9 \text{ m}$ $u_1 = u_0 = 0.55 \text{ m/s}$ $\beta = 30^\circ$

To find: Inclination angle θ
 Velocity of mass at point B.

i.e. ~~direct~~ angle between
 direction of v_B and horizontal
 plane at O-O

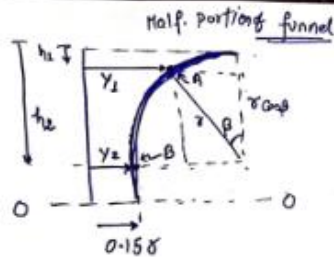


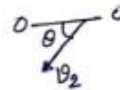
Fig. 1 Half portion of funnel

Step 1 : Apply Conservation of Angular momentum
 at point A and B.

3 marks

$$\therefore m u_1 y_1 = m u_2 y_2 \cos \theta$$

$$\Rightarrow \boxed{u_0 y_1 = u_2 y_2 \cos \theta} \quad \text{--- (1)}$$



Step 2 : Apply Conservation of Total mechanical energy (i.e. K.E & P.E)

3 marks

$$\frac{1}{2} m u_1^2 + m g (h_2 - h_1) = \frac{1}{2} m u_2^2$$

$$\begin{aligned} \Rightarrow u_2^2 &= u_1^2 + 2g(h_2 - h_1) \\ &= 0.55^2 + 2g(0.9 \cos \beta) \\ &= 15.59477 \end{aligned}$$

$$\begin{cases} r = 0.9 \text{ m} \\ \beta = 30^\circ \\ g = 9.81 \text{ m/s}^2 \end{cases}$$

$$\therefore \boxed{u_2 = 3.949 \text{ m/s}} \quad \text{--- (2)}$$

Step 3 : Calculation of y_1 .

2 marks

from Geometry of Fig. 1

$$\begin{aligned} y_1 &= y_2 + (r - r \sin \beta) \\ &= 0.15r + r(1 - \sin 30^\circ) \end{aligned}$$

$$\boxed{y_1 = 0.65r} \quad \text{--- (3)}$$

Step 4 Calculation of v_2 ~~from~~ Using Eq. (1), (2) and (3) as

Substituting $y_1 = 0.658$ into eq. (1)

$$\Rightarrow v_0 \times 0.658 = v_2 y_2 \cos \theta$$

$$\Rightarrow 0.55 \times 0.658 = 3.949 \times 0.15 \cos \theta$$

$$\Rightarrow \cos \theta = \frac{650}{1077}$$

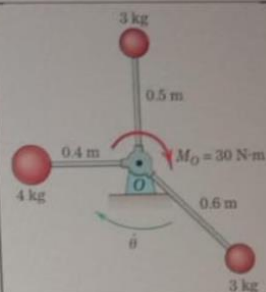
$$\Rightarrow \boxed{\theta = 52.877^\circ} \text{ Ans}$$

$$\approx 52.9^\circ \leftarrow \text{Book Answer}$$

Q 05

The three small spheres are welded to the light rigid frame which is rotating in a horizontal plane about a vertical axis through O with an angular velocity $\dot{\theta} = 20$ rad/s. If a couple $M_0 = 30$ N·m is applied to the frame for 5 seconds, compute the new angular velocity $\dot{\theta}'$.

Ans. $\dot{\theta}' = 80.7$ rad/s



Q5

10 marks

$$\boxed{\Sigma M_O = \dot{H}_O = \frac{dH_O}{dt}}, \int \Sigma M_O dt = \Delta H_O$$

$$M_0 t = \Delta |\Sigma m_i r_i (\dot{r}_i \dot{\theta})| = \Sigma m_i r_i^2 \Delta \dot{\theta}$$

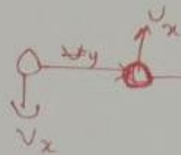
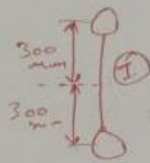
$$30 \times 5 = [3(0.5)^2 + 4(0.4)^2 + 3(0.6)^2](\dot{\theta}' - 20)$$

$$150 = 2.47(\dot{\theta}' - 20), \dot{\theta}' = 60.7 + 20 = 80.7 \frac{\text{rad}}{\text{s}}$$

10 marks

Q06

Question No 6.



F.B.D. 1 marks

Equation for impulse force along the horizontal axis

$$\int \Sigma F_s dt = 0 \quad \text{2.5 marks}$$

if we take ΔI_y as the impulse moment along y-axis.

$$\int \Sigma F_s dt = \Delta I_y$$

then $\Delta I_y = 2mV_y$

So

$$V_y = \frac{\Delta I_y}{2m} = \frac{10}{2 \times 1.5} = 3.3 \text{ m/s.}$$

$$\Delta I = 10 \text{ N.s}$$

$$m = 1.5 \text{ kg.}$$

2 marks

The sum of moment about the point I

$$\int \Sigma M_I dt = \Delta H_I \quad \text{2.5 marks}$$

$$\Delta H_I = I \times \text{dist.}$$

$$= 3 \text{ N.m.}$$

$$\Delta H_I = 2m \cdot V_x \times 0.3$$

$$V_x = 3.3 \text{ m/s.}$$

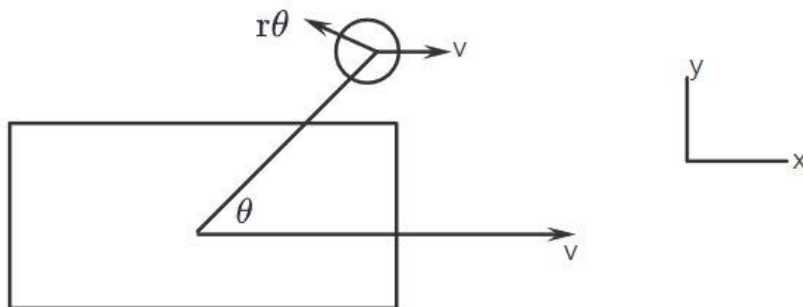
So Velocity of both spheres, $V = \sqrt{V_x^2 + V_y^2}$

$$V = \sqrt{3.3^2 + 3.3^2}$$

2 marks

$$V = 4.71 \text{ m/s}$$

Q7



the angular velocity of the sphere is $v = r\dot{\theta} = 0.4 \times 4 = 1.6 \text{ m/s}$

initial momentum of the system at $\theta = 0^\circ$ is given as ,

$$\Delta P_1 = (20 + 5) \times 0.6 = 15 \text{ Ns} \quad \text{---} \quad \text{3}$$

final momentum of the system at $\theta = 60^\circ$ is given as ,

$$\Delta P_2 = (20 + 5)v - 5(1.6)\sin 60^\circ \quad \text{---} \quad \text{3}$$

$$\Delta P_2 = 25v - 6.9282 \text{ N.s}$$

since there is no external force is applied on the system and there is no friction present so the momentum remains conserved.

$$\Delta P_1 = \Delta P_2 \quad \text{---} \quad \text{1}$$

$$15 = 25v - 6.9282$$

$$v = \frac{21.9282}{25} = 0.877128 \text{ m/s} \quad \text{---} \quad \text{3}$$